

CS23532
COMPUTER NETWORKS – LAB
PRACTICAL - 2
STUDY OF DIFFERENT TYPES OF NETWORK CABLES

AIM

To study different types of network cables, understand their construction and characteristics, compare their advantages and limitations, and learn how to make and test Ethernet straight-through and crossover cables.

1. Introduction

A network cable is a physical communication medium used to carry data between computers, switches, routers, access points and other network devices. Different cables are designed for different transmission speeds, distances, environments and applications.

The main cable families studied in this practical are twisted-pair cable, coaxial cable and fibre-optic cable. Additional modern cable types such as DAC and AOC are also included for broader understanding.

1.1 Objectives

- Identify the major types of network cables used in computer networks.
- Understand the construction and working principle of UTP, STP, coaxial and fibre-optic cables.
- Compare cable categories according to speed, distance, cost and applications.
- Understand the difference between straight-through and crossover Ethernet cables.
- Learn the T568A and T568B colour codes.
- Understand the procedure for terminating an Ethernet cable with an RJ45 plug.
- Test the completed cable and identify common wiring faults.

1.2 Major Types of Network Cables

Cable Type	Signal	Main Feature	Typical Application
UTP	Electrical	Low cost and easy installation	Ethernet LANS
STP/Shielded twisted pair	Electrical	Better protection from interference	Industrial/high-interference LANS
Coaxial	Electrical	Shielded single-conductor construction	TV, broadband, CCTV, RF
Fibre optic	Light	High bandwidth and EMI immunity	Backbone, data centres, telecom
DAC/Twinax	Electrical	Short high-speed interconnect	Data centres
AOC	Optical	Pre-terminated optical interconnect	

2. Twisted-Pair Cable

Twisted-pair cable consists of insulated copper conductors arranged in pairs. The conductors are twisted around each other to reduce electromagnetic interference and crosstalk. Ethernet cabling normally contains four twisted pairs.

2.1 Unshielded Twisted Pair (UTP)

UTP has no metallic shield around the individual pairs or around the complete cable. It is widely used because it is inexpensive, flexible and easy to install.

- **Advantages:** Low cost. Easy to install and terminate. Flexible and lightweight. Widely supported by Ethernet equipment. Suitable for many office and laboratory LAN installations.
- **Disadvantages:** More affected by electromagnetic interference than shielded cable. Cable performance depends strongly on correct installation and termination. Copper Ethernet has a limited channel distance.

2.2 Shielded Twisted Pair (STP)

STP uses metallic shielding to reduce the effect of external interference. Shielding can be applied around individual pairs, around groups of pairs, or around the complete cable depending on the cable construction.

- Better resistance to interference and noise.
- Useful in electrically noisy environments.
- Generally more expensive than UTP.
- Requires greater care during installation and grounding.

2.3 Other Shielding Forms

Type	Meaning/Construction	Main Benefit
F/UTP or FTP	Overall foil shield around the twisted pairs	Improved protection from external interference
S/UTP	Overall braided shield with unshielded pairs	Additional overall shielding
S/FTP	Overall braided shield plus foil around individual pairs	Strong shielding and pair isolation

Supplemental Note: Power over Ethernet (PoE)

UTP and STP cables do not just transmit data; they can also transmit electrical power. PoE is widely used to power devices like VoIP phones, security cameras, and wireless access points directly through the Ethernet cable, eliminating the need for a separate electrical outlet.

Supplemental Note: Cable Shielding and Safety Ratings

When dealing with Ethernet in a practical environment, the physical jacket of the cable matters.

- **Plenum vs. PVC:** Cables run through ceilings or walls with air circulation must have a **Plenum-rated** jacket. Standard PVC jackets release highly toxic smoke if they catch fire, whereas Plenum cables are coated with fire-retardant materials like Teflon.
- **Solid vs. Stranded Core:** The twisted pairs inside the cable can be solid copper (better for long, permanent wall runs as it is less prone to signal loss) or stranded copper (more flexible, better for the patch cables connecting your PC to the wall socket).

3. Ethernet Cable Categories

Cable category indicates the electrical performance of the cable. The actual network speed also depends on the Ethernet standard, cable length, connectors, installation quality and network devices.

Category	Common Speed/Use	Typical Reach	Notes
Cat3	10 Mbps/10BASE-T	Up to 100 m	Older Ethernet category
Cat5	100 Mbps; older installations may support 1 Gbps	Up to 100 m	Largely replaced by Cat5e
Cat5e	1 Gbps; supports some 2.5/5 Gbps systems	Up to 100 m	Very common for Gigabit Ethernet
Cat6	1 Gbps; 10 Gbps for shorter links	Up to 100 m depending on speed	Better performance margin than Cat5e
Cat6A	10 Gbps	Up to 100 m	Common choice for 10GBASE-T
Cat7	10 Gbps class	Up to 100 m in suitable systems	Usually shielded; higher-performance cabling class
Cat8	25/40 Gbps class	Typically up to 30 m	Mainly high-speed data-centre copper

4. Coaxial Cable

Coaxial cable has a central copper conductor, an insulating dielectric, a metallic shield and an outer protective jacket. The shield provides good resistance to external interference and helps control the electrical characteristics of the cable.

Type	Typical Use	Advantages	Limitations
RG-6	Cable TV, satellite and broadband	Good shielding and RF performance	Bulkier than twisted pair

RG-59	Video and CCTV applications	Flexible and widely used for video	Higher loss than some larger coax constructions
RG-11	Longer RF/video runs	Lower loss than smaller coax types	Thicker and less flexible

5. Fibre-Optic Cable

Fibre-optic cable transmits information using pulses of light through a glass or plastic optical medium. Unlike copper, fibre is not affected by electromagnetic interference and provides electrical isolation.

Type	Basic Characteristic	Advantages	Typical Applications
Single-mode fibre (SMF)	Small core; one main propagation mode	Long distance, high bandwidth, low attenuation	Telecom, backbone, long links
Multimode fibre (MMF)	Larger core; multiple propagation modes	Cost-effective for shorter optical links	Data centres, building/campus links

5.1 Advantages of Fibre

- Very high bandwidth.
- Long transmission distances.
- Immune to electromagnetic interference.
- Electrical isolation between network devices.
- Suitable for backbone and high-speed links.

5.2 Limitations of Fibre

- Higher installation and equipment cost.
- Termination and testing require specialised tools.
- Fibre must be handled carefully to avoid damage or contamination.

Supplemental Note: Specific Distances for Copper vs. Fiber

- **Copper Limit:** 100 meters (328 feet) is the strict maximum distance for **UTP/STP copper cables** (Cat5e, Cat6) before signal degradation (attenuation) occurs.
- **Fiber Optic Limit:** Fiber optic cables use light instead of electricity, allowing them to span vastly greater distances. **Multi-mode** fiber can generally reach between 300 to 550 meters, while **Single-mode** fiber (which uses a focused laser) can transmit data across distances ranging from 10 kilometers to over 100 kilometers without losing the signal.

6. Comparison of Major Cable Types

Parameter	UTP/STP	Coaxial	Fibre
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Signal	Electrical	Electrical	Light
Bandwidth	High for modern categories	Good for RF/broadband	Very high
EMI immunity	UTP: lower; STP: better	Good	Excellent
Installation	Easy	Moderate	More specialised
Cost	Low to moderate	Moderate	Higher
Long-distance use	Limited	Moderate	Excellent
Common use	LAN access	TV/RF/video	Backbone/data centre/telecom

7. Straight-Through and Crossover Ethernet Cables

The conductors inside an Ethernet cable are terminated according to a wiring standard. The two standards commonly studied are T568A and T568B.

7.1 Straight-Through Cable

A straight-through cable uses the same wiring standard at both ends. Therefore, the cable can be terminated as T568A-T568A or T568B-T568B.

7.2 Crossover Cable

A traditional crossover cable uses T568A at one end and T568B at the other. The transmit and receive pairs are therefore crossed.

Cable	End A	End B	Traditional Example
Straight-through	T568A	T568A	Same wiring standard at both ends
Straight-through	T568B	T568B	Common Ethernet patch-cable arrangement
Crossover	T568A	T568B	Traditional PC-to-PC or switch-to-switch

MODERN NETWORK NOTE: Many modern Ethernet interfaces support Auto-MDI/MDI-X. Because of this, a traditional crossover cable is often not required for direct device-to-device connections. The traditional distinction is still important for this laboratory practical.

Supplemental Note: Specific Device Connection Rules

- **Straight-Through Cables:** Used to connect **unlike** devices. For example: a PC to a Switch, or a Switch to a Router.
- **Crossover Cables:** Used to connect **like** devices. For example: a PC directly to another PC, a Switch to another Switch, or a Router directly to a PC.
- **Auto-MDIX Feature:** Modern Gigabit Ethernet network cards feature Auto-MDIX (Automatic Medium-Dependent Interface Crossover). This technology automatically detects the required cable connection

type and configures the connection appropriately, rendering physical crossover cables largely obsolete in modern networks.

8. T568A and T568B Colour Codes

Pin	T568A	T568B
1	White/Green	White/Orange
2	Green	Orange
3	White/Orange	White/Green
4	Blue	Blue
5	White/Blue	White/Blue
6	Orange	Green
7	White/Brown	White/Brown
8	Brown	Brown

MEMORY TIP: T568A begins with the green pair. T568B begins with the orange pair. The green and orange pairs are the two pairs that change positions between A and B.

9. Tools and Materials Required

- Cat5e or Cat6 Ethernet cable.
- Two RJ45/8P8C modular plugs.
- Crimping tool.
- Cable stripper and cutter.
- Cable tester (recommended).
- Cable boot/plug cover if required.
- T568A/T568B colour-code reference.

10. Procedure for Making an Ethernet Cable

1. Thread the cable boot onto the cable if a boot is being used.
2. Strip about 1-1.5 cm of the outer jacket carefully without damaging the conductors.
3. Separate and untwist the four twisted pairs only as much as necessary.
4. Choose the required wiring arrangement: T568A or T568B.
5. Arrange all eight conductors in the correct colour order.
6. Straighten the conductors and trim their ends evenly.
7. Insert the conductors completely into the RJ45 plug while maintaining the correct order.

8. Check that the outer jacket enters the plug far enough for the strain-relief section to grip it.
9. Place the plug in the crimping tool and crimp it firmly.
10. For a straight-through cable, repeat the same wiring standard at the second end.
11. For a traditional crossover cable, use the opposite standard at the second end.
12. Test the completed cable using a cable tester before using it in the network.

11. Cable Testing and Troubleshooting

Tester Result	Meaning	Likely Problem
1-1, 2-2, 3-3... 8-8	Correct straight-through mapping	Cable is wired correctly
Correct A-to-B crossover mapping	Correct crossover	Cable is wired as intended
Open/missing pin	No continuity	Broken wire or poor crimp
Two pins connected incorrectly	Short/miswire	Stray conductor or incorrect insertion
Wrong colour order	Miswire	Incorrect T568A/T568B sequence
Intermittent connection	Unreliable contact	Poor crimp, damaged plug or cable

12. Student Observation

Record the actual observations made during the laboratory session. The cable category should be identified from the markings printed on the cable jacket.

Observation	Record
Cable category used	
Cable type	UTP/STP/FTP/Other:
Number of twisted pairs	
Connector used	RJ45/8P8C
End A wiring	T568A/T568B
End B wiring	T568A/T568B
Cable made	Straight-through / Crossover
Cable tester result	Pass / Fail
Network test	Connected/Not connected

13. Answers to the Manual's Student Observation Questions

1. What is the difference between a cross cable and a straight cable?

A straight-through cable uses the same wiring standard at both ends, such as T568B at both ends. A traditional crossover cable uses T568A at one end and T568B at the other, crossing the relevant transmit and receive pairs.

2. Which type of cable is used to connect two PCs?

Traditionally, a crossover cable is used for a direct PC-to-PC Ethernet connection. However, modern network interfaces with Auto-MDI/MDI-X can often use a straight-through cable as well.

3. Which cable is used to connect a router/switch to your PC?

Traditionally, a straight-through Ethernet cable is used for PC-to-switch and PC-to-router connections.

4. Find the category of twisted-pair cable used in your lab.

Read the marking printed on the cable jacket. It may be marked CAT5E, CAT6, CAT6A or another category. The category should be recorded from the actual cable used in the laboratory.

5. Write your understanding, challenges faced and output received.

Sample observation: I understood how the four twisted pairs are arranged and terminated in an RJ45 plug. The main challenge was maintaining the correct colour sequence while inserting the conductors into the connector. After crimping, the cable was tested with a cable tester. The cable tester showed the expected wire mapping and the completed cable was suitable for the intended connection.

14. Advantages and Disadvantages - Quick Revision

Cable	Advantages	Disadvantages
UTP	Low cost, flexible, easy to install	Less protection from EMI
STP	Better protection from interference	More expensive; installation/grounding needs care
Coaxial	Good shielding and RF performance	Bulkier and less common for modern LAN access
Fibre	High bandwidth, long distance, EMI immune	More expensive and requires specialised installation
DAC	Low latency and economical for short links	Short reach
AOC	High speed and light weight	More costly than passive copper and has fixed ends